

# RES-5 / GAP-2 arc consolidation: the matched-order-to-exact problem reduces

TECT verification-first repository · claim B1-RH-ENUM

first issued 2026-06-10 · this version issued 2026-06-10 · v1.0

## 1. Purpose and scope

RES-5 (= GAP-2) is the conversion of the matched-second-cumulant Reading-H selection into a statement about EXACT free energies. This note consolidates the full arc into a single record: the reduction, the corrections caught en route, and the honest final status. No new theorem. Scope: B1’s lattice T6 scope; the  $I = 2 \times 10^{-3}$  endpoint.

## 2. The arc

| step        | note                                 | result                                                                       |
|-------------|--------------------------------------|------------------------------------------------------------------------------|
| entry       | res5/gap2-entry                      | remainder common-mode suppressed by $a_0$ (T3)                               |
| bare route  | susceptibility-ratio-bareroute       | ELIMINATED (NG); bare ratio 9.05                                             |
| framing     | oneloop-disentangling                | one-loop is exact; RES-5 is the LOOP expansion                               |
| skeleton    | higherloop-commonmode                | higher-loop = 2PI skeleton expansion                                         |
| response    | a0-skeleton-sensitivity              | screened $C_G = 1/(1 + g) = 0.49$ (no amplification)                         |
| 2PI         | 2pi-commonmode-monotonicity          | R-U10-3 extends to the 2PI action                                            |
| certificate | sunset-norm v1.0→ <b>v1.1</b>        | <b>survival RETRACTED</b> ; correction kept; RES-5 OPEN                      |
| tail budget | tail-budget-closure                  | localised to the $I = 2 \times 10^{-3}$ endpoint                             |
| endpoint    | endpoint-2pi-bound v1.0→ <b>v1.2</b> | bracketed; <b>SC-SCOPE framing corrected</b>                                 |
| projection  | projection-factor-bound              | <b>NEGATIVE</b> $\chi_{\text{proj}} = 1.25$ ; tail $0.047 \rightarrow 0.059$ |
| DR-2        | dr2-kappa-bound v1.0→ <b>v1.2</b>    | EXACT(enum)/SE(lattice)/T6-cond(arb- $Q$ ); <b>“uncond.” downgraded</b>      |
| RES-1       | hdiag-offdiag-floor-bound            | H-diag parallels RES-5 ( $R_{\text{lead}} < 1$ , same $K_{\text{floor}}$ )   |
| pin         | endpoint-admissible-pin              | worst $K_{\text{floor}} = 12.13$ (unif) / $T' = 13$ (non-unif)               |

Three corrections (bold) were caught en route: the v1.0 “survives” over-claim, the v1.1 SC-SCOPE-as-named-hypothesis framing, and the v1.1 “unconditional T7” grade – each retracted in the ledger.

## 3. The reduction

Every line converged on one object: the additive-energy floor

$$K_{\text{floor}}(Q) = \frac{E_+(Q)}{N^2} - 1, \quad E_+(Q) = \sum_t w_t^2, \quad w_t = \sum_{u+v=t} A_u A_v.$$

The endpoint closes (corrected tail 0.059) iff  $K_{\text{floor}} < 26.2$ ; H-diag/RES-1 (off-diagonal Bogoliubov stability,  $R_{\text{lead}} = \text{const}(1 + K_{\text{floor}})I$ ) iff  $K_{\text{floor}} < 20.5$ . The off-endpoint ( $I \leq 10^{-3}$ ) regime is strong-evidence closed with a  $\geq 27\times$  margin; only the  $I = 2 \times 10^{-3}$  endpoint is binding.

## 4. Final status

| class                                                | max $K_{\text{floor}}$                                                      | status                                                                              |
|------------------------------------------------------|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| enumerated / admissible-pinned<br>full lattice class | 12.13 (unif), $T' = 13$ (non-unif)<br>$C_\varepsilon R^\varepsilon$ (R-026) | <b>EXACT</b> ( $< 20.5 < 26.2$ )<br><b>STRONG EVIDENCE</b> (T7<br>mod. textbook NT) |
| arbitrary- $Q$ (not B1 scope)                        | $N^\varepsilon$ (decoupling)                                                | <b>T6 CONDITIONAL</b><br>(Bourgain–Demeter)                                         |

For the actual Reading-H competitors the RES-5 endpoint and H-diag close EXACTLY (the additive-energy floor is pinned  $\leq 12.13$ , a factor  $\geq 1.58$  below the binding threshold), for uniform AND non-uniform amplitudes. So RES-5 is NOT an independent B1 blocker. The lattice-class theorem is strong evidence (R-026); the arbitrary- $Q$  case is conditional on decoupling and lies outside B1’s lattice scope.

## 5. The single residual; tiers

The one remaining item is the carrier-richness link

$$\chi(P) \lesssim T'(Q),$$

from the additive energy  $E_+/T'$  to the actual STEP-5B layer floor – a B5 operator-decision, not the additive-energy bound itself (which is pinned). Closing it makes the RES-5 endpoint and H-diag theorems over the admissible lattice class. No tier flip: B1-RH-ENUM and B2-PROPA-HLAYER stay T6 CONDITIONAL on {H-LAYER}. Within B1’s scope the H-LAYER residual is now: RES-5 EXACT/strong-evidence (this arc) + RES-1 parallel + the isotropy-infimum theorem (Math427); the deepest open piece is the  $\chi(P)$  link and the full off-diagonal operator norm.

## 6. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- ( $\alpha$ ) “*A consolidation can quietly upgrade tiers.*” DISMISSED: no tier is changed; the final table restates the per-note grades (EXACT / STRONG EVIDENCE / T6-cond), and the three retracted over-claims are recorded, not buried.
- ( $\beta$ ) “*Is ‘EXACT for enumerated’ a theorem?*” The additive-energy floor  $K_{\text{floor}} \leq 12.13$  is an EXACT combinatorial computation over the admissible class; the closure THRESHOLD (20.5/26.2) is estimate-grade and the  $\chi(P)$  link is open – so the endpoint is EXACT-pinned at the additive-energy level, strong-evidence at the physical-floor level. The wording is kept precise.
- ( $\gamma$ ) “*What did the arc actually achieve?*” It reduced RES-5 (the matched-order-to-exact problem) from an open analytic frontier to a pinned additive- energy floor sharing one machinery with H-diag, H-ADM-COH, and DR-2; localised the only stress to the  $I = 2 \times 10^{-3}$  endpoint; corrected the tail ( $0.047 \rightarrow 0.059$ ); and identified the single residual ( $\chi(P)$  link). B1’s H-LAYER residual is correspondingly narrowed.

Quantitative sanity check (mandatory): *pin*  $K_{\text{floor}} \leq 12.13 < 20.5 < 26.2$ ; *margin*  $\geq 1.58$ ; *tail* corrected  $0.047 \rightarrow 0.059$ ; *off-endpoint*  $\geq 27\times$ ; *corrections* 3 retracted (survival / SC-SCOPE / unconditional-T7).

## 7. Result footer

Result ID: B1-RH-ENUM / RES-5 arc consolidation

Precise statement: RES-5 (matched-order-to-exact) reduces to the additive-energy floor  $K_{\text{floor}} = E_+ / N^2 - 1$ . Endpoint closes iff  $K_{\text{floor}} < 26.2$ ; H-diag/RES-1 iff  $K_{\text{floor}} < 20.5$ ; exact worst over the admissible crystallographic-shell class = 12.13 (uniform) /  $T' = 13$  (non-uniform, R-027), both below threshold (factor  $\geq 1.58$ ). EXACT for the actual competitors (not an independent B1 blocker); STRONG EVIDENCE for the full lattice class (R-026, T7 mod. textbook NT); T6-cond on decoupling for arbitrary-Q. Single residual:  $\chi(P) < \sim T'$

link. Tail corrected  $0.047 \rightarrow 0.059$  (projection negative). Three over-claims retracted en route.

Scope: B1 lattice T6 scope;  $I = 2e-3$  endpoint; consolidation (no new theorem).

Dependencies: res5-{entry,susceptibility-ratio,oneloop-disentangling,higherloop,a0-skeleton,2pi-commonmode,sunset-norm v1.1,tail-budget,endpoint v1.2,projection-factor,dr2-kappa v1.2}; hdiag-offdiag-floor-bound; res-endpoint-admissible-pin; R-026/R-027; B5 status.json.

Evidence grade: CONSOLIDATION of EXACT + STRONG EVIDENCE + T6-cond material; no new numbers beyond the cited per-step scripts.

Reproduction command: python codes/vacuum/res\_endpoint\_admissible\_pin.py

Expected output: worst  $K_{\text{floor}}$  12.13 (uniform) /  $T' = 13$  (non-uniform);  $< 20.5 < 26.2$ ; 4/4 PASS (plus the cited per-step scripts).

Falsification gate: An admissible competitor with  $K_{\text{floor}} > 20.5$ ; or the  $\chi(P) < \sim T'$  link failing; or R-026 failing over the admissible range.

Tier before / after: No tier action. B1/B2 T6 CONDITIONAL on {H-LAYER}. RES-5/GAP-2: EXACT (enumerated/pinned) / STRONG EVIDENCE (lattice) / T6-cond (arbitrary-Q). Not an independent B1 blocker.

No-overclaim statement: Consolidation only. EXACT is at the additive-energy level (combinatorial pin); the physical-floor closure is STRONG EVIDENCE (estimate-grade threshold + open  $\chi(P)$  link); the lattice theorem is R-026 (T7 mod. textbook NT); arbitrary-Q is T6-cond on decoupling. Three over-claims were retracted in the

ledger. B1/B2 T6 unchanged.  
 Next required action:   Close the  $\chi(P) \leftarrow T'$  carrier-richness link (B5  
                           operator-decision)  
                           + the full off-diagonal operator norm (RES-1)  
                           to make RES-5 +  
                           H-diag theorems over the admissible lattice  
                           class. The deepest  
                           remaining H-LAYER piece is then the  
                           isotropy-infimum / Prop-A core.